

Frontend theory Assignment-2

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1. What is CSS Box Model?

The CSS Box Model describes how every HTML element is structured. It consists of:

Content → Padding → Border → Margin

2. What is Media Query?

A media query is used to make a website **responsive** for different screen sizes.

```
@media (max-width: 600px) {  
  body  
{  
  background-color: lightblue;  
}  
}
```

3. Features of Bootstrap

- Responsive design
- Ready-made components
- Grid system
- Easy to use
- Mobile-friendly
- Cross-browser support

4. Basic Grid Structure in Bootstrap

Bootstrap uses a **12-column grid system**.

```
<div class="container">  
  
  <div class="row">  
  
    <div class="col-6">Column 1</div>  
  
    <div class="col-6">Column 2</div>  
  
  </div>  
  
</div>
```

5. What are CSS Selectors?

CSS selectors are used to **select HTML elements** for applying styles.

Examples:

- Element: p
- Class: .box
- ID: #header
- Universal: *

6. CSS Backgrounds and Properties

CSS backgrounds are used to set the background of an element.

Main properties:

- background-color
- background-image
- background-repeat
- background-position
- background-size
- background-attachment

7. How can we overlap elements in CSS?

We can overlap elements using **positioning and z-index**.

```
.box {  
  position: absolute;  
  z-index: 2;  
}
```

8. Different Positioning Properties in CSS

The main positioning properties are:

- static
- relative
- absolute
- fixed
- sticky

9. What is ECMAScript?

ECMAScript is the standard/specification on which JavaScript is based. It defines JavaScript features and syntax.

10. Difference between var, let and const + Temporal Dead Zone

var	let	const
Function scoped	Block scoped	Block scoped
Can be redeclared	Cannot be redeclared	Cannot be redeclared
Can be reassigned	Can be reassigned	Cannot be reassigned

Temporal Dead Zone (TDZ):

The TDZ is the period between entering a block and declaring a let or const variable. During this time, the variable **cannot be accessed**.

```
console.log(x); // Error
```

```
let x = 10;
```